EDG results

Number of trials = 50.

Synthetic Data Setup

```
p = 200;  % number of points
d = 2;    % dimension of the points
hs = haltonset(d, 'Skip', 1e3, 'Leap', 1e2);  % Skip initial values and leap
every 100 points to reduce correlation
hs = scramble(hs, 'RR2');  % Scramble for more randomness
P1 = -100 + 200 * net(hs, m);
P2 = -100+200.*rand(d,n);
P = [P1' P2];
```

A new set of points was created in each trial. Here, I used Halton sequence to generate anchor points that are more evenly distributed across a given range.

Mean

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	1.6644	4.8773	7.4582	10.7870	13.9533	16.8563
15	0.5007	1.2537	3.5453	5.5559	7.6791	10.2648
20	0.0000	0.0512	1.1366	2.0057	3.5153	5.6019
25	0.0000	0.0000	0.1561	0.5925	1.3811	3.0230
30	0.0000	0.0000	0.0393	0.2838	0.6991	1.5715
35	0.0000	0.0000	0.0000	0.0086	0.2845	0.7048
40	0.0000	0.0000	0.0000	0.0093	0.0000	0.0963

Median

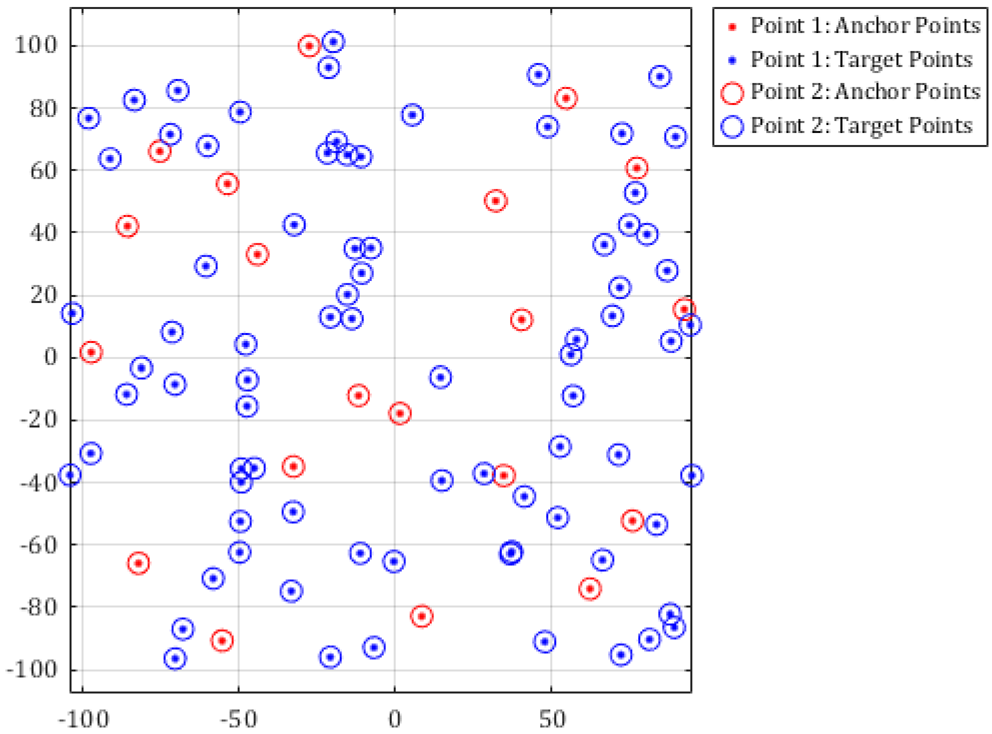
m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	2.3117	4.6129	7.6105	10.9890	13.5476	16.8235
15	0.0000	1.6858	3.4340	5.3425	7.4408	10.0010
20	0.0000	0.0000	0.9324	1.9321	3.6257	5.4719

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
25	0.0000	0.0000	0.0000	0.0000	1.7235	3.2005
30	0.0000	0.0000	0.0000	0.0000	0.0000	1.6872
35	0.0000	0.0000	0.0000	0.0000	0.0000	0.0373
40	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Mean with std

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	1.8015 (1.1544)	4.4768 (1.3605)	7.9265 (1.4675)	10.6125 (1.3052)	13.8051 (1.4085)	16.8680 (1.6278)
15	0.3791 (0.7636)	1.6319 (1.2650)	3.4107 (0.9617)	5.0982 (1.2948)	7.7705 (1.3568)	10.1639 (1.4287)
20	0.0000 (0.0000)	0.2525 (0.6169)	0.4871 (0.7688)	2.0924 (1.1136)	3.7899 (1.3831)	5.5610 (1.0960)
25	0.0000 (0.0000)	0.0000 (0.0000)	0.3087 (0.6924)	0.6397 (1.0371)	1.4652 (1.0332)	2.8230 (1.1224)
30	0.0000 (0.0000)	0.0000 (0.0000)	0.0780 (0.3944)	0.0912 (0.2907)	0.5660 (0.8589)	1.5715 (1.1667)
35	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0555 (0.2527)	0.0943 (0.3004)	0.4866 (0.7083)
40	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0274 (0.1870)	0.0837 (0.4210)	0.1915 (0.5132)

Sample plot for 100 points



Protein 1ubq

Results (Mean)

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	10.2331	13.2233	15.1545	16.9073	18.9403	19.9589
15	7.2271	8.7707	10.4461	11.6341	11.6932	13.2317
20	3.1518	6.4313	7.5524	8.0298	9.1087	9.8561
25	1.8750	2.6563	4.1939	5.6223	6.6341	7.5699
30	0.1311	1.7871	2.4159	3.9703	5.1380	6.0758
35	0.0375	1.0781	1.5369	1.7231	2.4865	3.2119
40	0.0003	0.0387	0.1437	0.7253	1.3741	2.1563

Results (median)

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	10.3143	13.4958	15.8351	17.4581	18.7076	19.6969
15	6.5315	9.2369	10.3983	11.0058	12.1442	12.9566
20	2.1976	6.5284	7.3957	8.5619	9.1254	9.9854

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
25	1.4894	2.2717	4.2312	6.1905	6.3833	7.2428
30	0.0000	1.6604	2.1755	3.0234	5.7921	6.1635
35	0.0000	0.9514	1.3201	1.5852	2.0021	3.1194
40	0.0000	0.0000	0.0540	0.5402	1.4131	1.9861

Mean with std

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	10.2479 (1.0474)	13.1742 (1.6311)	16.1357 (1.7920)	17.2658 (2.0215)	19.4230 (2.6291)	21.0285 (2.8794)
15	7.0081 (1.5983)	8.7383 (2.0079)	10.7667 (1.4529)	11.6421 (1.6956)	13.0368 (2.6234)	13.2605 (2.3018)
20	2.9133 (1.6362)	6.4989 (1.7401)	7.2842 (1.2059)	8.0339 (1.1759)	8.9292 (1.0005)	9.6885 (1.0737)
25	1.4678 (0.7909)	2.8652 (1.7320)	4.4908 (2.1681)	5.9097 (1.2287)	6.2118 (0.7398)	7.3919 (0.7928)
30	0.1312 (0.4943)	1.8197 (1.2363)	2.7187 (1.3359)	4.0938 (1.6400)	5.0885 (1.1467)	5.9277 (0.7843)
35	0.0461 (0.2313)	1.2767 (0.9545)	1.4749 (0.6614)	1.8041 (0.7033)	2.5138 (1.1211)	3.2360 (1.0090)
40	0.0000 (0.0000)	0.0145 (0.0506)	0.2046 (0.2169)	0.6086 (0.5977)	1.5480 (0.7665)	2.2299 (0.5692)

Protein 1w2e

Mean with std

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
10	44.9008 (6.4650)	62.1561 (12.5342)	73.6129 (13.3319)	85.4721 (17.8514)	96.8980 (17.0431)	102.8494 (19.6843)
20	16.9671 (1.9896)	18.2897 (2.6545)	22.6780 (3.1816)	26.1954 (4.3822)	26.5001 (4.5147)	31.9628 (5.5458)

m \ alpha	0.05	0.10	0.15	0.20	0.25	0.30
30	14.4225 (2.8534)	13.9136 (1.4326)	14.7908 (1.5348)	16.5525 (2.1708)	16.6837 (2.1208)	18.2033 (2.7335)
40	7.7945 (3.2760)	8.3098 (1.0406)	9.4697 (2.0128)	10.4827 (2.0423)	11.2159 (1.8299)	11.9114 (1.9545)
50	2.6309 (2.4151)	6.1718 (3.0983)	7.3430 (1.5980)	7.3104 (1.4763)	8.3036 (0.8547)	8.4584 (1.0388)
60	1.2012 (0.4219)	1.6834 (1.0047)	2.4478 (2.2114)	3.7762 (3.2112)	5.8510 (3.4403)	7.9401 (2.5983)
70	1.5011 (0.3050)	1.7089 (0.3994)	1.9036 (0.5586)	3.2721 (2.3776)	3.8977 (2.6602)	5.1123 (2.9083)
80	1.0714 (0.7251)	1.5367 (0.4660)	1.7788 (0.4535)	1.9302 (0.8560)	2.2275 (1.3797)	2.5412 (1.4644)
90	0.0713 (0.3546)	0.7414 (1.0439)	1.8692 (0.6039)	1.9124 (0.5898)	2.0029 (0.4667)	2.1494 (0.4500)
100	0.0000 (0.0000)	0.0917 (0.4547)	0.6443 (1.2651)	1.1014 (1.3561)	1.8325 (1.4999)	1.8145 (1.4339)
110	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0038 (0.0268)	0.1087 (0.3830)	0.1941 (0.4975)